

USAF ACADEMY  
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING  
LAB #6 (Lesson 21) - Array Factor Lab  
**Lab Sheet**  
Fall 2026

**Name:** \_\_\_\_\_ **Section:** \_\_\_\_\_ **Date:** \_\_\_\_\_

**Learning Objective 3.2:** I can derive the array factor for an arbitrary linear array and apply pattern multiplication.

This sheet is the turn-in document for the Lesson 21 lab. Fill the calculated columns before you sweep and the measured columns at the bench, so the prediction is on paper before the trace is on the screen. The lab is a team assignment, so one sheet per team.

- 1. The measurement table.** One row per aperture, at 10.3 GHz with  $d = 14$  mm. Report each trace's peak level, and the step from the eight-element peak beside it, and report each sidelobe level relative to that trace's own peak.

| Active elements | Peak (dBFS) | HPBW meas | HPBW calc | FNBW meas | FNBW calc | First SLL |
|-----------------|-------------|-----------|-----------|-----------|-----------|-----------|
| 8 (Rx1-Rx8)     |             |           |           |           |           |           |
| 4 (Rx3-Rx6)     |             |           |           |           |           |           |
| 2 (Rx4, Rx5)    |             |           |           |           |           |           |

- 2. Written answer 1.** Explain why the first-null beamwidth of the two-element array is quoted as  $180^\circ$  rather than measured. Name the condition on  $\lambda/2d$  that decides whether a null exists in visible space, and evaluate it for  $d = 14$  mm at 10.3 GHz.

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- 3. Written answer 2.** The gain of each patch element is the same whether the other seven elements are on or off, and the amplifier behind each element is unchanged. Explain why the peak of the sweep nonetheless drops by about 6 dB every time you halve the number of active elements.

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**Documentation:** \_\_\_\_\_